# Time Series Project: IPI Food Industries

```r
knitr::opts_chunk$set(
  echo = TRUE,
  warning = TRUE,
  message = TRUE,
  fig.align = "center",
  fig.width = 10,
  fig.height = 6
)

required_pkgs <- c(
  "readr", "dplyr", "ggplot2", "lubridate",
  "forecast", "tseries", "ellipse", "knitr"
)

missing_pkgs <- required_pkgs[!vapply(required_pkgs, requireNamespace, logical(1), quietly = TRUE)]
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
if (length(missing_pkgs) > 0) {
  install.packages(missing_pkgs, repos = "https://cloud.r-project.org")
}

library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(tseries)
library(forecast)
library(knitr)

d <- 1
```

# 1 Part I: The data

## 1.1 1. What does the chosen series represent?

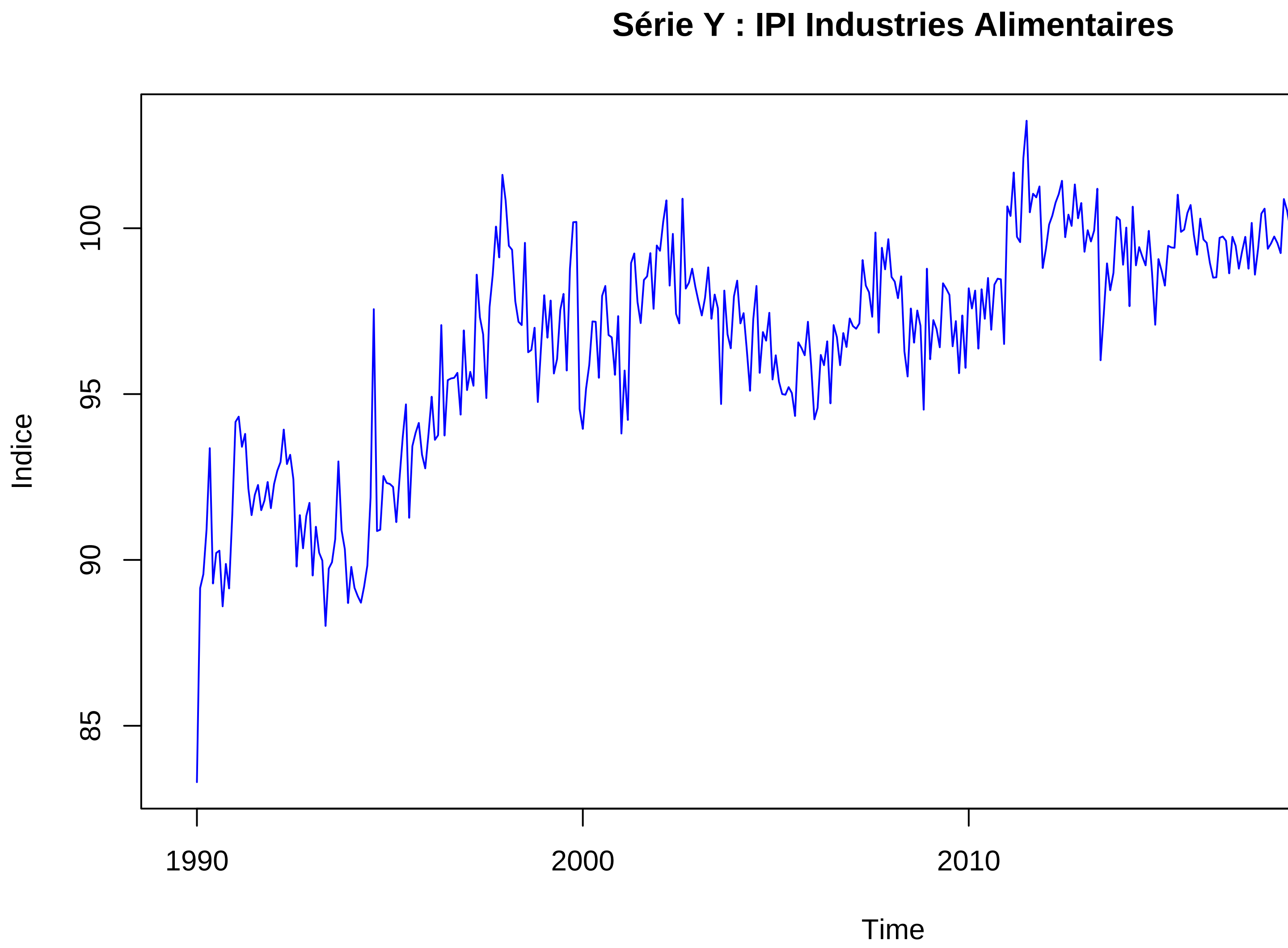
The dataset is an Industrial Production Index (IPI) for the food industries sector in France, with monthly frequency, seasonally and working-day adjusted (CVS-CJO), based at 100 in 2021. The variable is an index and is unit-free. Since the series is already CVS-CJO, we do not apply additional seasonal adjustment.

**Modelling context.**

- As an index (based at 100 in 2021), values above (below) 100 indicate production above (below) its 2021 average level for the sector.
- The CVS-CJO correction removes most deterministic seasonal patterns and calendar effects; consequently, we do not expect pronounced remaining seasonality and we focus on non-seasonal ARIMA/ARMA dynamics.
- Because the index is strictly positive over the sample, the analysis can be carried out on $Z_t = \log(Y_t)$; first differences of $Z_t$ have a growth-rate interpretation and often provide variance stabilization.

## 1.2 2 and 3. Transform the series to make it stationary if necessary whith graphical representation before and after transformation (placed here for convenience of the study)

```
df <- read.csv("data/valeurs_mensuelles.csv", sep = ";", skip = 4, header = FALSE, stringsAsFactors = FA

df <- df[, 1:2]
colnames(df) <- c("date_brute", "valeur")

df <- df[df$valeur != "" & !is.na(df$valeur), ]

# On ajoute "-01" pour créer un format AAAA-MM-JJ lisible par R
df$date <- as.Date(paste0(df$date_brute, "-01"))
df <- df[order(df$date), ]

# On récupère l'année et le mois de début
start_y <- as.numeric(format(df$date[1], "%Y"))
start_m <- as.numeric(format(df$date[1], "%m"))

Y <- ts(df$valeur, start = c(start_y, start_m), frequency = 12)

plot(Y, main = "Série Y : IPI Industries Alimentaires", ylab = "Indice", col = "blue")
```

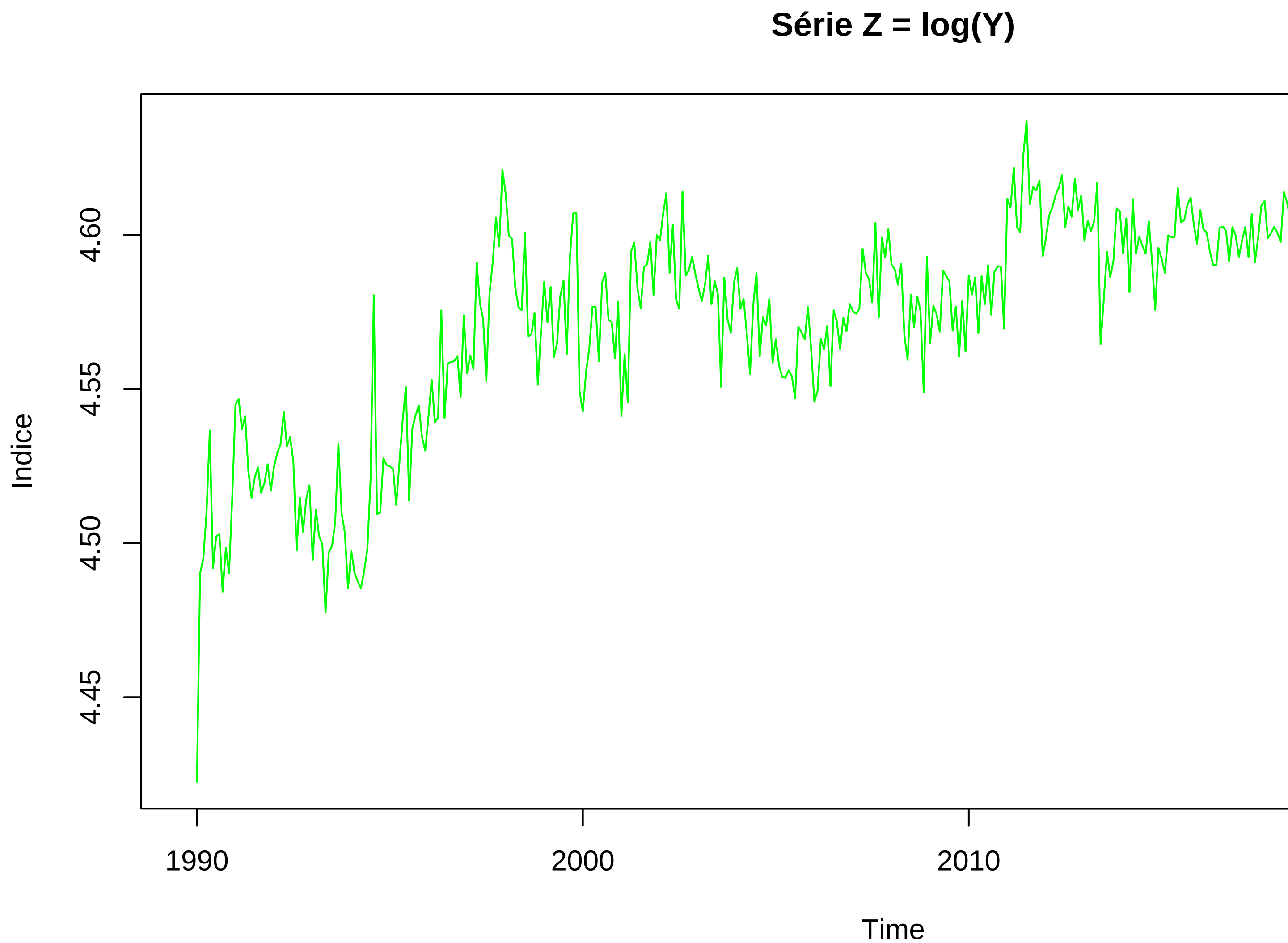



Visual inspection of the raw series $Y_t$ shows that the amplitude of fluctuations increases proportionally with the level of the index. To address this heteroscedasticity, we apply a natural logarithm transformation:

$$Z_t = \ln(Y_t) \tag{1}$$

**Justification:** This transformation stabilizes the variance and converts multiplicative growth patterns into additive ones. Economically, the first difference of the logged series approximates the monthly growth rate, a standard metric for industrial production analysis.

```
Z <- log(Y)
plot(Z, main = "Série Z = log(Y)", ylab = "Indice", col = "green")
```

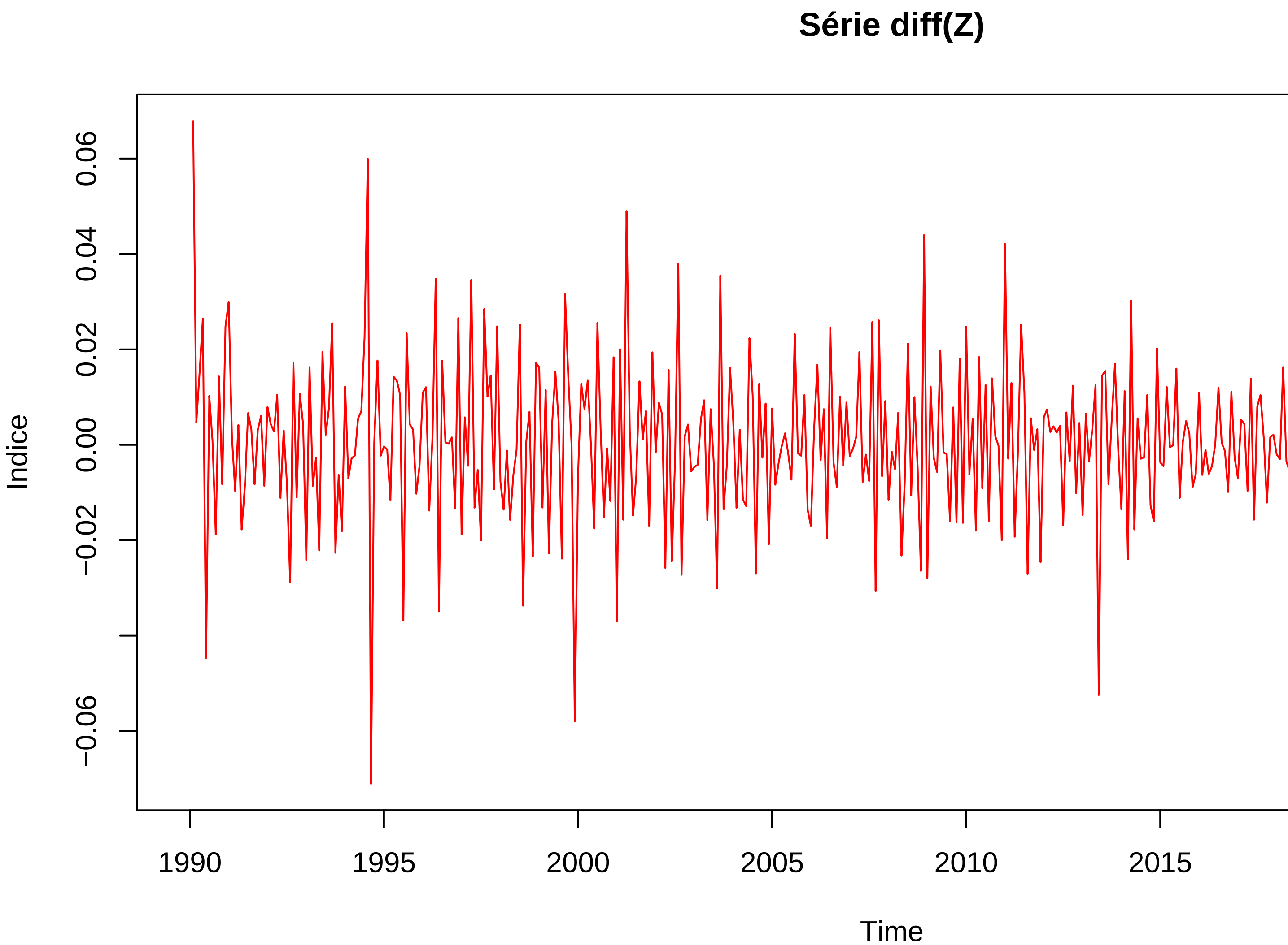



The series $Z_t$ exhibits a clear upward long-term trend, indicating that the mean is not constant over time. To remove this stochastic trend and eliminate the unit root, we apply first-order differencing $(d = 1)$:

$$X_t = \Delta Z_t = \ln(Y_t) - \ln(Y_{t-1} = \nabla^1 \log(Y_t)) \tag{2}$$

**Justification:** This process transforms the series into a "Difference-Stationary" process. The resulting series $X_t$ represents the monthly innovations in production, which is a requirement for identifying ARMA $(p, q)$ processes.

```
X <- diff(Z, differences = d)
plot(X, main = "Série diff(Z)", ylab = "Indice", col = "red")
```




To rigorously confirm that the transformed series $X_t$ is stationary, we performed the following tests based on the processed data:

```
adf.test(X)
```

```
## Warning in adf.test(X): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  X
## Dickey-Fuller = -11.349, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```r
kpss.test(X)
```

```
## Warning in kpss.test(X): p-value greater than printed p-value
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  X
## KPSS Level = 0.18423, Truncation lag parameter = 5, p-value = 0.1
```

The Dickey-Fuller statistic (-11.349) is significantly lower than the standard critical values (e.g., -2.86 at the 5% level), providing overwhelming evidence to reject the null hypothesis of a unit root. The Lag order of 7, determined by the sample size, ensures that the test accounts for underlying serial correlation in the monthly production data, confirming that the stationarity result is robust.

The KPSS test confirms the results of the ADF test. The test statistic (KPSS Level = 0.18423) is well below the 5% critical value of 0.463. With a p-value of 0.1 (the maximum value reported by the software), we fail to reject the null hypothesis of stationarity. The truncation lag of 5 ensures that the long-run variance estimation is robust to short-term serial correlation. Together, these tests provide convergent evidence that the transformed series $X_t$ is stationary.

# 2 Part II: ARMA models

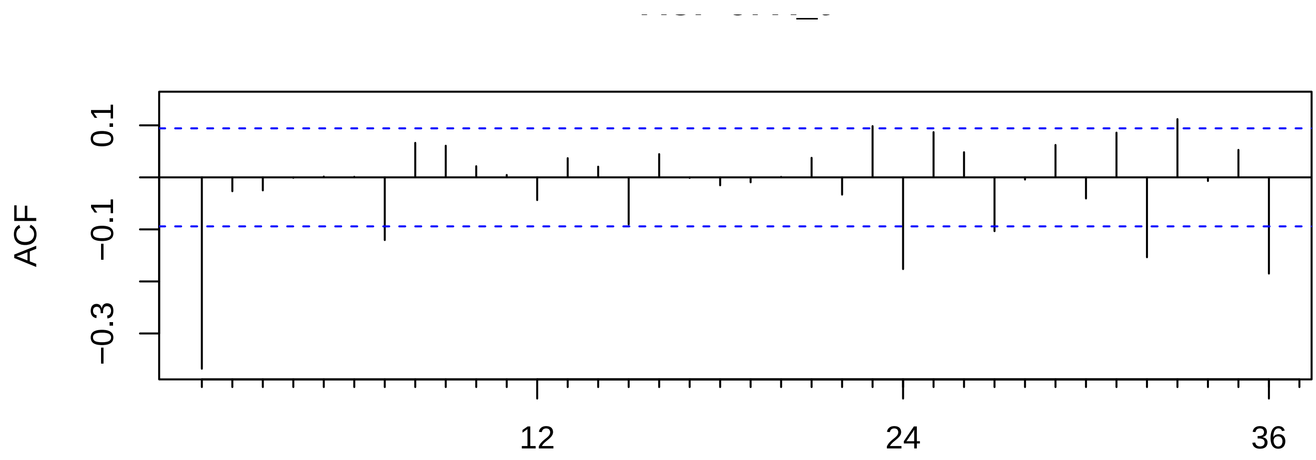
## 2.1 4. Pick and estimate an ARMA(p,q) model for $X_t$

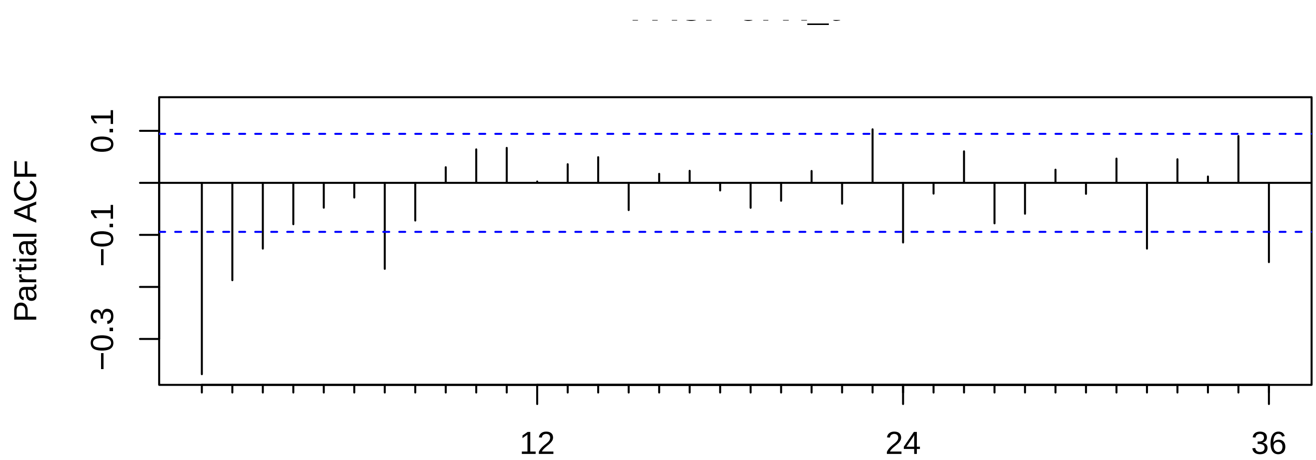
We use ACF/PACF diagnostics and an information-criterion search on a small grid to select a parsimonious ARMA model.

**Model identification and selection.**

- **Identification logic.** The ACF/PACF of $X_t$ are used as qualitative guidance: short-range dependence suggests an ARMA with relatively small orders rather than a high-order AR (pure PACF cutoff) or pure MA (pure ACF cutoff).
- **Quantitative selection.** We then compare ARMA$(p, q)$ candidates on a grid $(0 \leq p, q \leq 5)$ and select the feasible model minimizing AIC. AIC is preferred here because the goal is forecasting/fit (it penalizes complexity less harshly than BIC).

```r
op <- par(mfrow = c(2, 1), mar = c(3, 4, 2, 1))
forecast::Acf(X, lag.max = 36, main = "ACF of X_t")
forecast::Pacf(X, lag.max = 36, main = "PACF of X_t")
```




```
par(op)
```

```
max_p <- 5
max_q <- 5

grid <- expand.grid(p = 0:max_p, q = 0:max_q) %>%
  rowwise() %>%
  mutate(
    fit = list(tryCatch(
      Arima(X, order = c(p, 0, q), include.mean = TRUE),
      error = function(e) NULL
    )),
    aic = ifelse(is.null(fit), NA, AIC(fit)),
    bic = ifelse(is.null(fit), NA, BIC(fit))
  ) %>%
  ungroup() %>%
  arrange(aic)

grid %>%
  filter(!is.na(aic)) %>%
  select(p, q, aic, bic) %>%
  head(5) %>%
```

```
  kable()
```

| p | q | aic | bic |
|---|---|---|---|
| 4 | 5 | -2477.383 | -2432.605 |
| 5 | 5 | -2475.192 | -2426.344 |
| 3 | 3 | -2472.856 | -2440.290 |
| 4 | 3 | -2471.407 | -2434.770 |
| 3 | 4 | -2471.403 | -2434.766 |

```
best_row <- grid %>% filter(!is.na(aic)) %>% slice(1)
best_fit <- best_row$fit[[1]]
best_row %>% select(p, q, aic, bic) %>% kable()
```

| p | q | aic | bic |
|---|---|---|---|
| 4 | 5 | -2477.383 | -2432.605 |

We select the model with the lowest AIC among feasible candidates, which yields ARMA($p = 4, q = 5$) for $X_t$. The estimated parameters are:

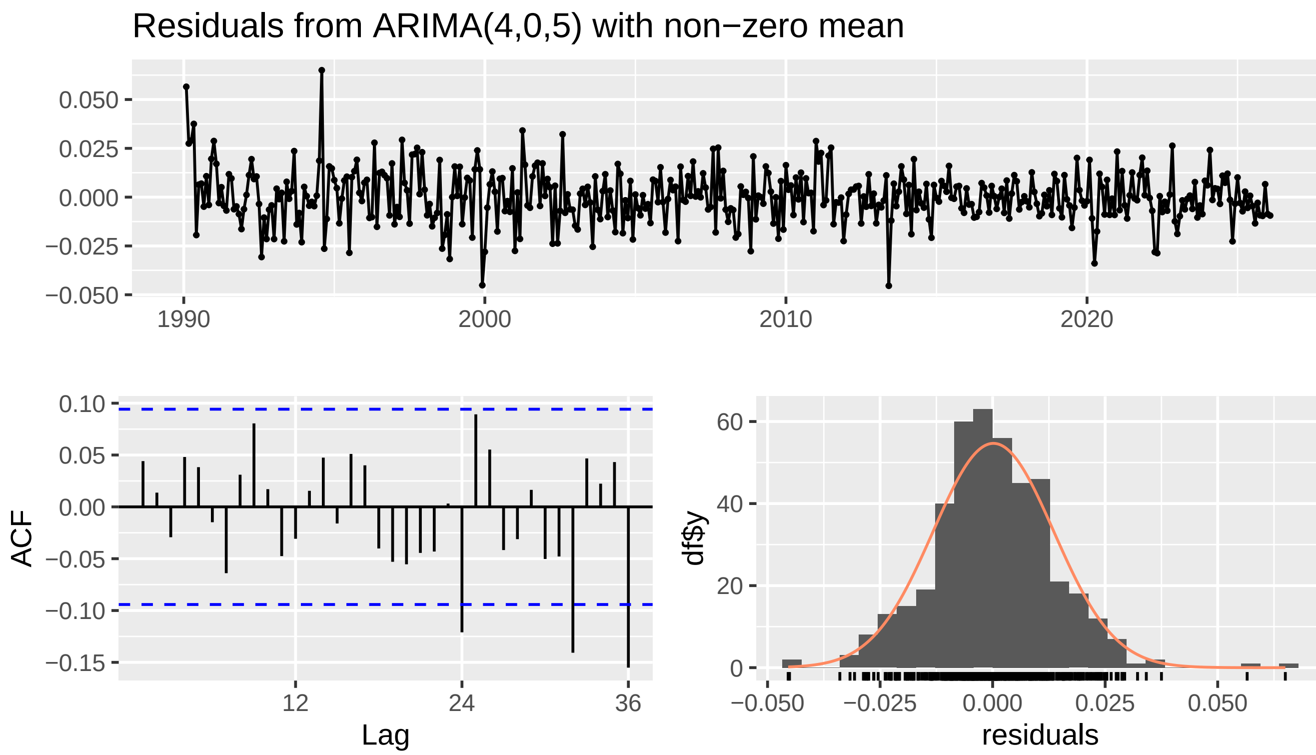
```
coef <- best_fit$coef
se <- sqrt(diag(best_fit$var.coef))
coef_tbl <- tibble::tibble(
  term = names(coef),
  estimate = unname(coef),
  std_error = unname(se),
  t_value = unname(coef / se)
) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

coef_tbl %>% kable()
```

| term | estimate | std_error | t_value |
|---|---|---|---|
| ar1 | 0.0540 | 0.0213 | 2.5353 |
| ar2 | 0.6039 | 0.0237 | 25.5144 |
| ar3 | 0.0133 | 0.0448 | 0.2964 |
| ar4 | -0.9645 | 0.0174 | -55.5530 |
| ma1 | -0.6251 | 0.0733 | -8.5317 |
| ma2 | -0.6444 | 0.0658 | -9.7895 |
| ma3 | 0.3518 | 0.1027 | 3.4252 |
| ma4 | 0.9960 | 0.0484 | 20.5728 |
| ma5 | -0.6083 | 0.1064 | -5.7159 |
| intercept | 0.0003 | 0.0002 | 1.0657 |

### 2.1.1 Diagnostic checks

We validate the fitted model via residual autocorrelation, Ljung-Box test, and a Gaussianity check.

```r
checkresiduals(best_fit, test = FALSE)
```




**Residual diagnostics.**

- **Whiteness.** The Ljung–Box test checks for remaining autocorrelation in residuals. A non-significant p-value supports the ARMA adequacy (no leftover linear dependence).
- **Gaussianity.** The Jarque–Bera test is a strict check for normality. For many macro series, residuals can be non-Gaussian because of outliers (e.g., crisis months), which affects distributional statements (prediction intervals/regions) more than point forecasts.
- **Here (numbers).** Ljung–Box at lag 24 (df adjusted) gives p-value 0.0528, and Jarque–Bera gives p-value $1.89 \times 10^{-15}$.

Overall, the model is considered acceptable for linear dependence if residual autocorrelation is not significant; the Gaussian assumption used later for the confidence region should be interpreted as an approximation if normality is rejected.

## 2.2 5. Write the ARIMA(p,d,q) model for the chosen series

Let $Z_t$ be the transformed series (log if used) and $X_t = \nabla^d Z_t$ be the stationary series. If the selected ARMA model is ARMA$(p,q)$ for $X_t$, then the original series follows an ARIMA$(p,d,q)$ model for $Z_t$. If $Z_t = \log Y_t$, this corresponds to an ARIMA$(p,d,q)$ for the log series, which implies a multiplicative structure on $Y_t$.

**ARIMA representation.** With $d = 1$ and the selected ARMA$(p = 4, q = 5)$ for $X_t = \nabla^d Z_t$, the model can be written as

$$\phi(B)\,\nabla^d Z_t = c + \theta(B)\,\varepsilon_t, \qquad \varepsilon_t \sim \text{i.i.d. } (0, \sigma^2),$$

where $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ and $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$. When $Z_t = \log(Y_t)$ and $d = 1$, a non-zero mean in $X_t$ corresponds to a **drift** in $\log(Y_t)$, i.e., a constant average growth component.

# 3 Part III: Prediction

Denote $T$ the length of the series. We assume the residuals are Gaussian.

## 3.1 6. Confidence region for $(X_{T+1}, X_{T+2})$

Let $\hat{\mu} = (\hat{X}_{T+1}, \hat{X}_{T+2})^\top$ be the 2-step-ahead forecast mean vector and $\hat{\Sigma}$ the 2x2 forecast error covariance matrix. Under Gaussian residuals, the joint forecast error is bivariate normal, so the level-$\alpha$ confidence region is the ellipse

$$\mathcal{C}_\alpha = \left\{ x \in \mathbb{R}^2 : (x - \hat{\mu})^\top \hat{\Sigma}^{-1} (x - \hat{\mu}) \leq \chi^2_{2,\alpha} \right\}.$$

For an ARMA model, let $\{\psi_j\}_{j \geq 0}$ be the coefficients of the MA($\infty$) representation. Then

- $\mathrm{Var}(X_{T+1} - \hat{X}_{T+1}) = \sigma^2$,
- $\mathrm{Var}(X_{T+2} - \hat{X}_{T+2}) = \sigma^2(1 + \psi_1^2)$,
- $\mathrm{Cov}(X_{T+1} - \hat{X}_{T+1}, X_{T+2} - \hat{X}_{T+2}) = \sigma^2 \psi_1$,

where $\sigma^2$ is the innovation variance.

**Interpretation.** This region is the multivariate analogue of a univariate prediction interval: it is a joint set such that $\mathbb{P}((X_{T+1}, X_{T+2}) \in \mathcal{C}_\alpha) \approx \alpha$ under the model. The chi-square threshold comes from the quadratic form of a bivariate normal vector. In practice, if residuals are heavier-tailed than Gaussian, the ellipse may be too small (undercoverage).

## 3.2 7. Hypotheses used to obtain this region

1. The ARMA model for $X_t$ is correctly specified.
2. Innovations are i.i.d. Gaussian with variance $\sigma^2$.
3. Model parameters are treated as known (estimation uncertainty ignored for the region).
4. The process is stationary and invertible (so the MA($\infty$) representation exists).

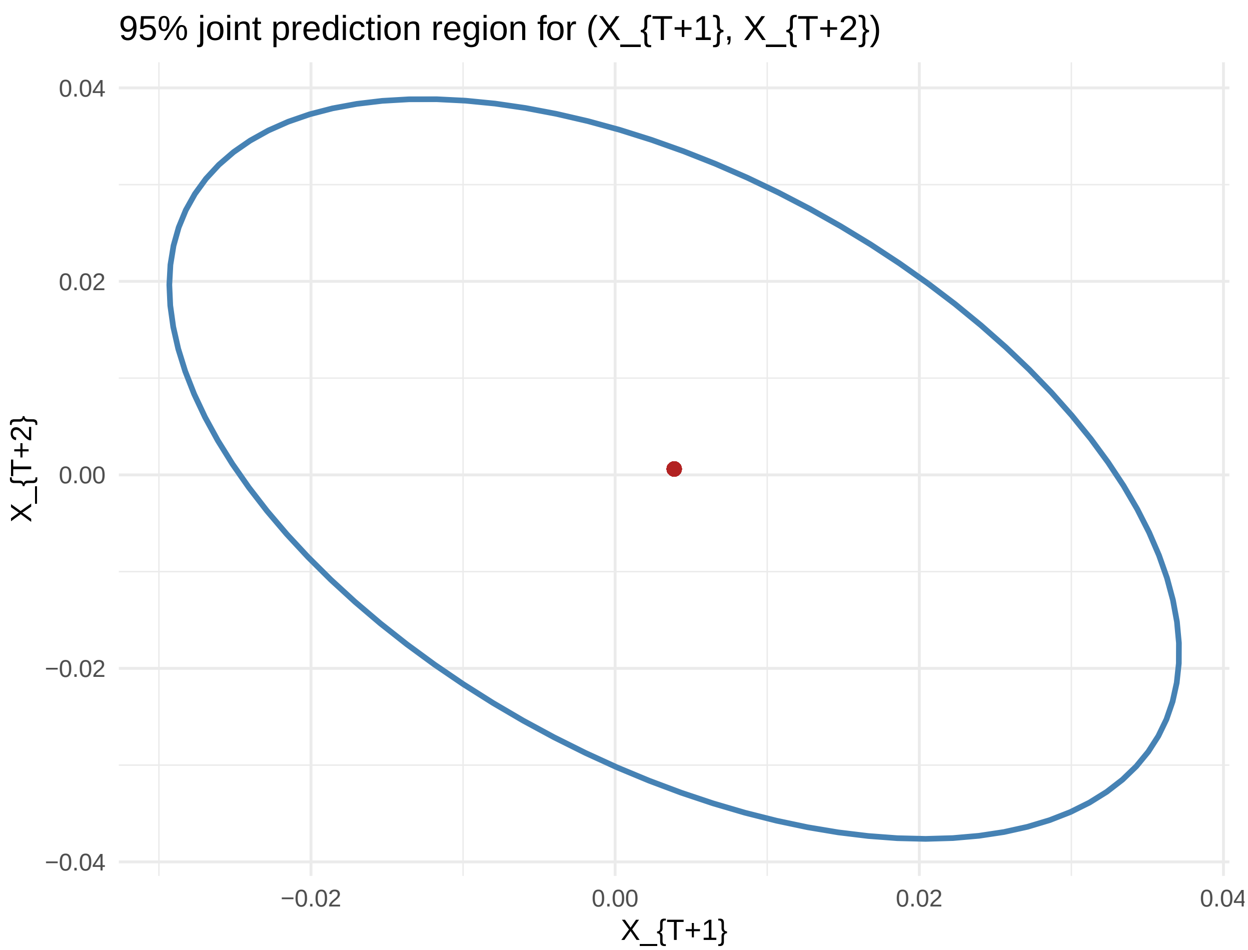
**Role of the assumptions.**

- If the ARMA is misspecified, $\hat{\Sigma}$ is wrong and the region is not calibrated.
- Gaussianity is what makes the contour exactly elliptical with a chi-square cutoff; without it, ellipses are at best an approximation.
- Treating parameters as known ignores estimation uncertainty and typically makes the region slightly too optimistic; this effect is non-negligible in small samples but tends to shrink as $T$ grows.

## 3.3 8. Graphical representation for $\alpha = 95\%$

```r
alpha <- 0.95

pred <- predict(best_fit, n.ahead = 2)
mu <- as.numeric(pred$pred)

# Extract AR and MA coefficients for psi1
ar <- if (!is.null(best_fit$model$phi)) best_fit$model$phi else numeric(0)
ma <- if (!is.null(best_fit$model$theta)) best_fit$model$theta else numeric(0)
psi1 <- if (length(ar) == 0 && length(ma) == 0) 0 else ARMAtoMA(ar, ma, lag.max = 1)[1]

sigma2 <- best_fit$sigma2
Sigma <- matrix(
  c(sigma2, sigma2 * psi1, sigma2 * psi1, sigma2 * (1 + psi1^2)),
  nrow = 2
)

ell <- as.data.frame(ellipse::ellipse(Sigma, centre = mu, level = alpha))
colnames(ell) <- c("X_T1", "X_T2")

corr12 <- Sigma[1, 2] / sqrt(Sigma[1, 1] * Sigma[2, 2])

p <- ggplot(ell, aes(x = X_T1, y = X_T2)) +
  geom_path(color = "steelblue", linewidth = 1) +
  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2) +
  labs(
    title = "95% joint prediction region for (X_{T+1}, X_{T+2})",
    x = "X_{T+1}",
    y = "X_{T+2}"
  ) +
  theme_minimal()

p
```

```
## Warning in geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2): All aesthetics have
## i Please consider using ‘annotate()‘ or provide this layer with data containing
##   a single row.
```




**Comments on the 95% region.**

- The forecast-error correlation is -0.496, i.e., **negative**: a positive error at horizon 1 tends to be associated with a negative error at horizon 2 under the fitted ARMA dynamics.
- Using $\rho = \psi_1/\sqrt{1+\psi_1^2}$, this corresponds to $\psi_1 \approx \rho/\sqrt{1-\rho^2}$, hence here $\psi_1 \approx -0.571$.
- The ellipse is therefore tilted downward; its size is governed by the innovation variance $\sigma^2 = 1.84 \times 10^{-4}$ (on the $X_t$ scale) and by $|\psi_1|$.

### 3.4 9. Open question

Let $Y_t$ be a stationary time series observed from $t = 1$ to $T$, and suppose $Y_{T+1}$ is observed before $X_{T+1}$. Using $Y_{T+1}$ improves the prediction of $X_{T+1}$ if and only if $Y_{T+1}$ contains incremental predictive information about $X_{T+1}$ beyond the sigma-field generated by $\{X_1, \ldots, X_T\}$. Formally, if

$$\mathbb{E}[X_{T+1} \mid X_1, \ldots, X_T, Y_{T+1}] \neq \mathbb{E}[X_{T+1} \mid X_1, \ldots, X_T],$$

then including $Y_{T+1}$ improves the forecast in mean-square error.

A practical test is to estimate an ARMAX-type model for $X_t$ with $Y_t$ (or a proxy for $Y_{T+1}$ if only available at $T+1$) and test the significance of the coefficient(s) on $Y$. This can be done via a t-test on the added regressor(s), or a likelihood-ratio test comparing the ARMA model for $X_t$ against the ARMAX model.

**Conditions and practical test.**

- A convenient equivalent condition is that $Y_{T+1}$ is not conditionally independent of $X_{T+1}$ given past $X$: informally, $Y$ must have **incremental predictive content** for $X$.
- In linear/Gaussian settings, this amounts to $Y$ being correlated with the one-step-ahead innovation (forecast error) of $X$ when conditioning on $\mathcal{F}_T = \sigma(X_1, \ldots, X_T)$.
- Practical implementation: fit an ARMAX model $X_t = \text{ARMA terms} + \beta Y_t + \varepsilon_t$ (or a lead/nowcast proxy for $Y_{T+1}$) and test $H_0 : \beta = 0$ (t-test / Wald) or compare log-likelihoods (LR test). A rejection indicates that observing $Y_{T+1}$ can reduce the MSE of forecasting $X_{T+1}$.

# 4 Appendix: Code

```
knitr::opts_chunk$set(

  echo = TRUE,

  warning = TRUE,

  message = TRUE,

  fig.align = "center",

  fig.width = 10,

  fig.height = 6

)



required_pkgs <- c(

  "readr", "dplyr", "ggplot2", "lubridate",

  "forecast", "tseries", "ellipse", "knitr"

)



missing_pkgs <- required_pkgs[!vapply(required_pkgs, requireNamespace, logical(1), quietly = TRUE)]

if (length(missing_pkgs) > 0) {

  install.packages(missing_pkgs, repos = "https://cloud.r-project.org")

}



library(dplyr)

library(ggplot2)
```

```r
library(tseries)
library(forecast)
library(knitr)

d <- 1

df <- read.csv("data/valeurs_mensuelles.csv", sep = ";", skip = 4, header = FALSE, stringsAsFactors = FA

df <- df[, 1:2]
colnames(df) <- c("date_brute", "valeur")

df <- df[df$valeur != "" & !is.na(df$valeur), ]

# On ajoute "-01" pour créer un format AAAA-MM-JJ lisible par R
df$date <- as.Date(paste0(df$date_brute, "-01"))
df <- df[order(df$date), ]

# On récupère l'année et le mois de début
start_y <- as.numeric(format(df$date[1], "%Y"))
start_m <- as.numeric(format(df$date[1], "%m"))

Y <- ts(df$valeur, start = c(start_y, start_m), frequency = 12)

plot(Y, main = "Série Y : IPI Industries Alimentaires", ylab = "Indice", col = "blue")
Z <- log(Y)
plot(Z, main = "Série Z = log(Y)", ylab = "Indice", col = "green")
X <- diff(Z, differences = d)
```

```r
plot(X, main = "Série diff(Z)", ylab = "Indice", col = "red")
adf.test(X)
kpss.test(X)
op <- par(mfrow = c(2, 1), mar = c(3, 4, 2, 1))
forecast::Acf(X, lag.max = 36, main = "ACF of X_t")
forecast::Pacf(X, lag.max = 36, main = "PACF of X_t")
par(op)
max_p <- 5
max_q <- 5

grid <- expand.grid(p = 0:max_p, q = 0:max_q) %>%
  rowwise() %>%
  mutate(
    fit = list(tryCatch(
      Arima(X, order = c(p, 0, q), include.mean = TRUE),
      error = function(e) NULL
    )),
    aic = ifelse(is.null(fit), NA, AIC(fit)),
    bic = ifelse(is.null(fit), NA, BIC(fit))
  ) %>%
  ungroup() %>%
  arrange(aic)

grid %>%
  filter(!is.na(aic)) %>%
  select(p, q, aic, bic) %>%
  head(5) %>%
```

```r
  kable()

best_row <- grid %>% filter(!is.na(aic)) %>% slice(1)
best_fit <- best_row$fit[[1]]
best_row %>% select(p, q, aic, bic) %>% kable()
coef <- best_fit$coef
se <- sqrt(diag(best_fit$var.coef))
coef_tbl <- tibble::tibble(
  term = names(coef),
  estimate = unname(coef),
  std_error = unname(se),
  t_value = unname(coef / se)
) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

coef_tbl %>% kable()
checkresiduals(best_fit, test = FALSE)
res <- residuals(best_fit)
lb_test <- Box.test(res, lag = 24, type = "Ljung-Box", fitdf = sum(best_fit$arma[1:4]))
jb_test <- tseries::jarque.bera.test(res)
alpha <- 0.95

pred <- predict(best_fit, n.ahead = 2)
mu <- as.numeric(pred$pred)

# Extract AR and MA coefficients for psi1
ar <- if (!is.null(best_fit$model$phi)) best_fit$model$phi else numeric(0)
```

```r
ma <- if (!is.null(best_fit$model$theta)) best_fit$model$theta else numeric(0)
psi1 <- if (length(ar) == 0 && length(ma) == 0) 0 else ARMAtoMA(ar, ma, lag.max = 1)[1]

sigma2 <- best_fit$sigma2
Sigma <- matrix(
  c(sigma2, sigma2 * psi1, sigma2 * psi1, sigma2 * (1 + psi1^2)),
  nrow = 2
)

ell <- as.data.frame(ellipse::ellipse(Sigma, centre = mu, level = alpha))
colnames(ell) <- c("X_T1", "X_T2")

corr12 <- Sigma[1, 2] / sqrt(Sigma[1, 1] * Sigma[2, 2])

p <- ggplot(ell, aes(x = X_T1, y = X_T2)) +
  geom_path(color = "steelblue", linewidth = 1) +
  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2) +
  labs(
    title = "95% joint prediction region for (X_{T+1}, X_{T+2})",
    x = "X_{T+1}",
    y = "X_{T+2}"
  ) +
  theme_minimal()

p
code_chunks <- knitr::knit_code$get()
code_text <- unlist(code_chunks, use.names = FALSE)
```

```r
cat("```r\n")

cat(paste(code_text, collapse = "\n\n"))

cat("\n```\n")
```